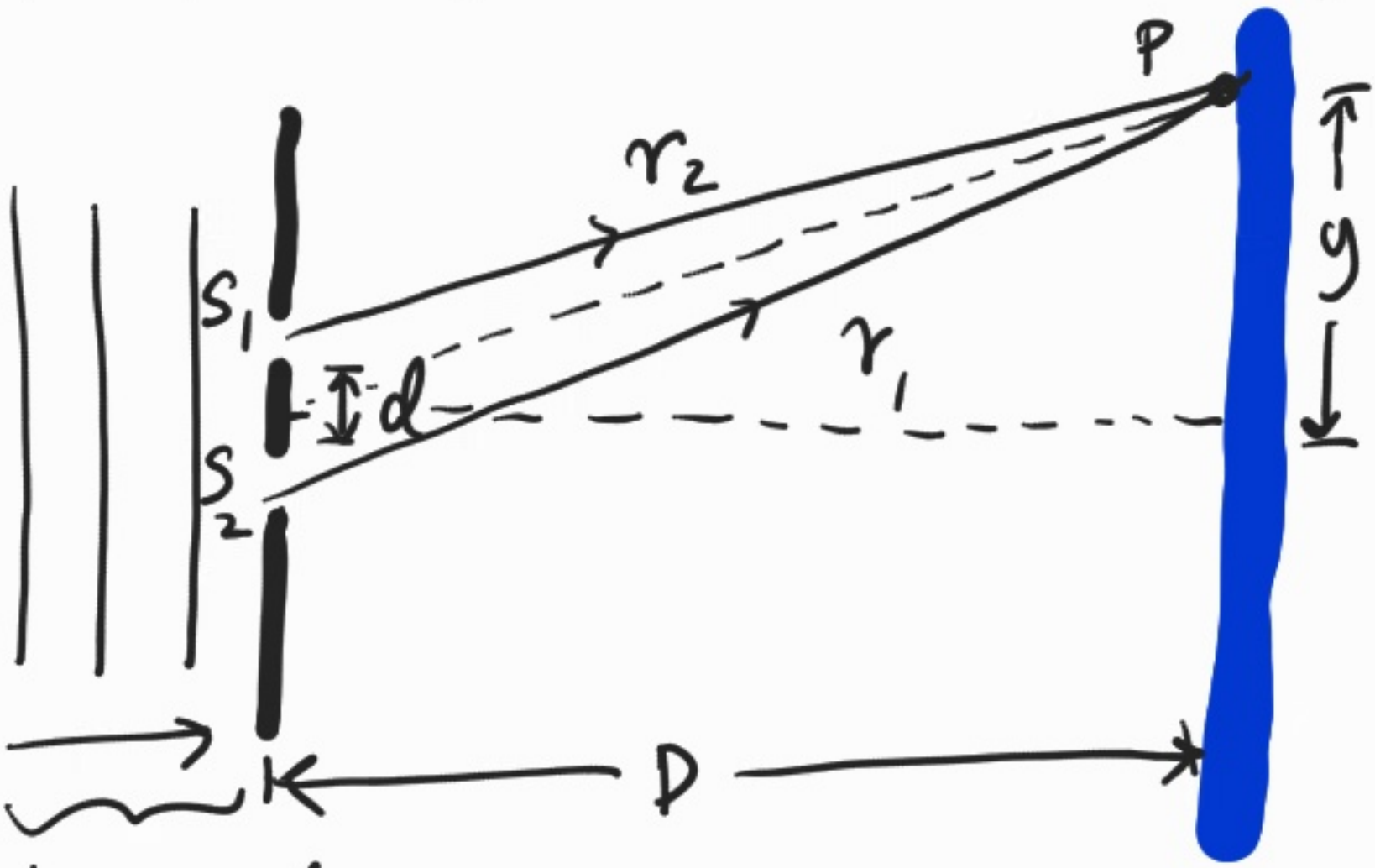
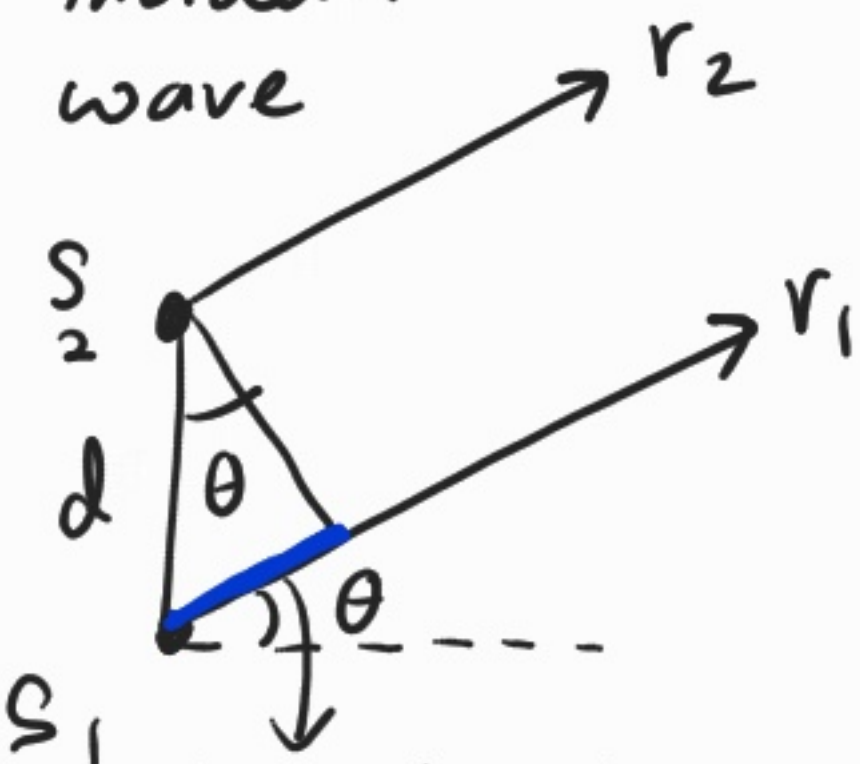
Young's Experiment

⇒ Double-slit interference experiment.



incident wave



Path length difference, ΔL } $\Delta L = d \sin\theta$

Young's Experiment

- [Constructive Interference]

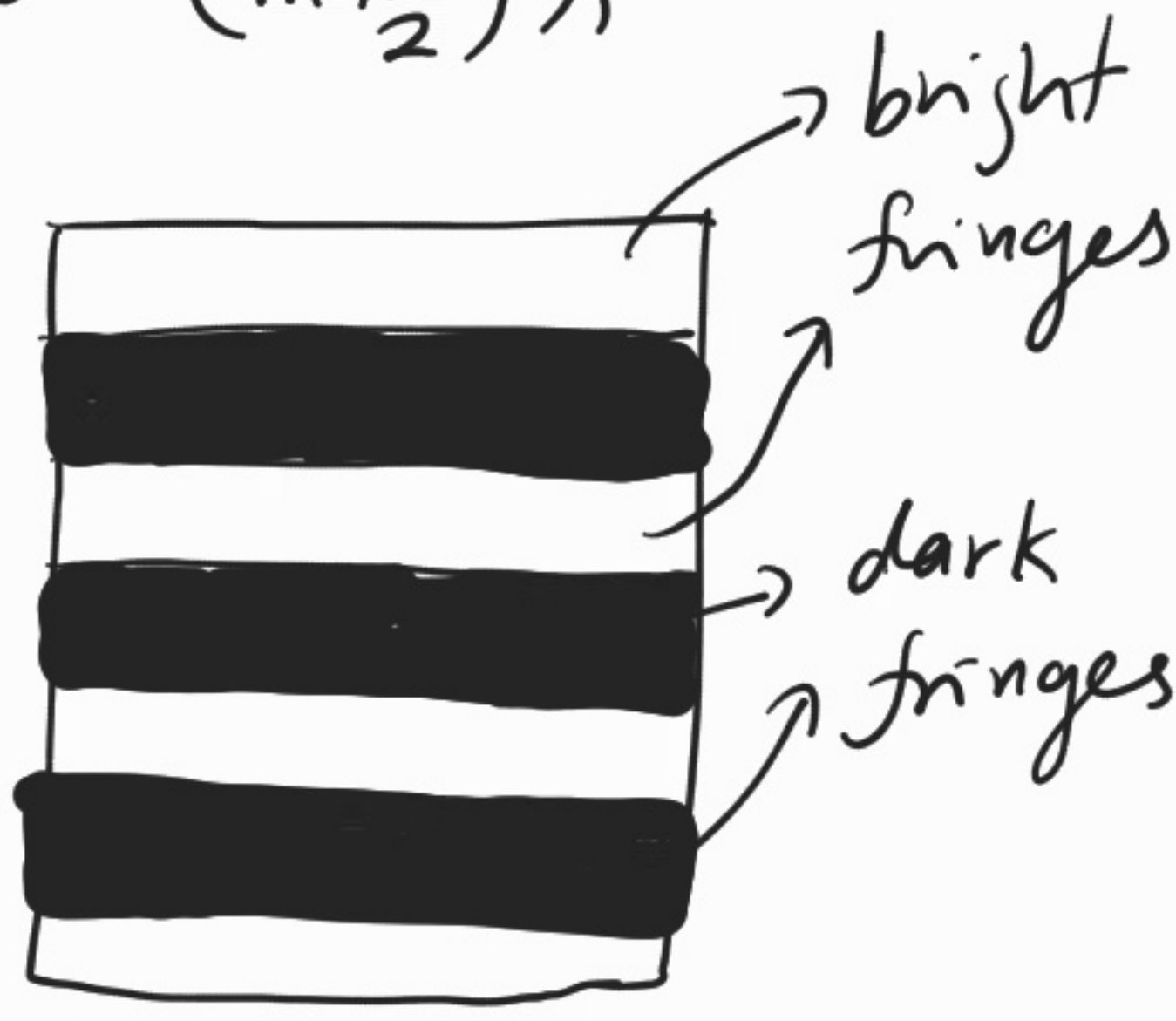
Max → Bright fringes

$d \sin\theta = m\lambda$

- [Destructive Interference]

Min → Dark fringes

$d \sin\theta = (m+\frac{1}{2})\lambda$

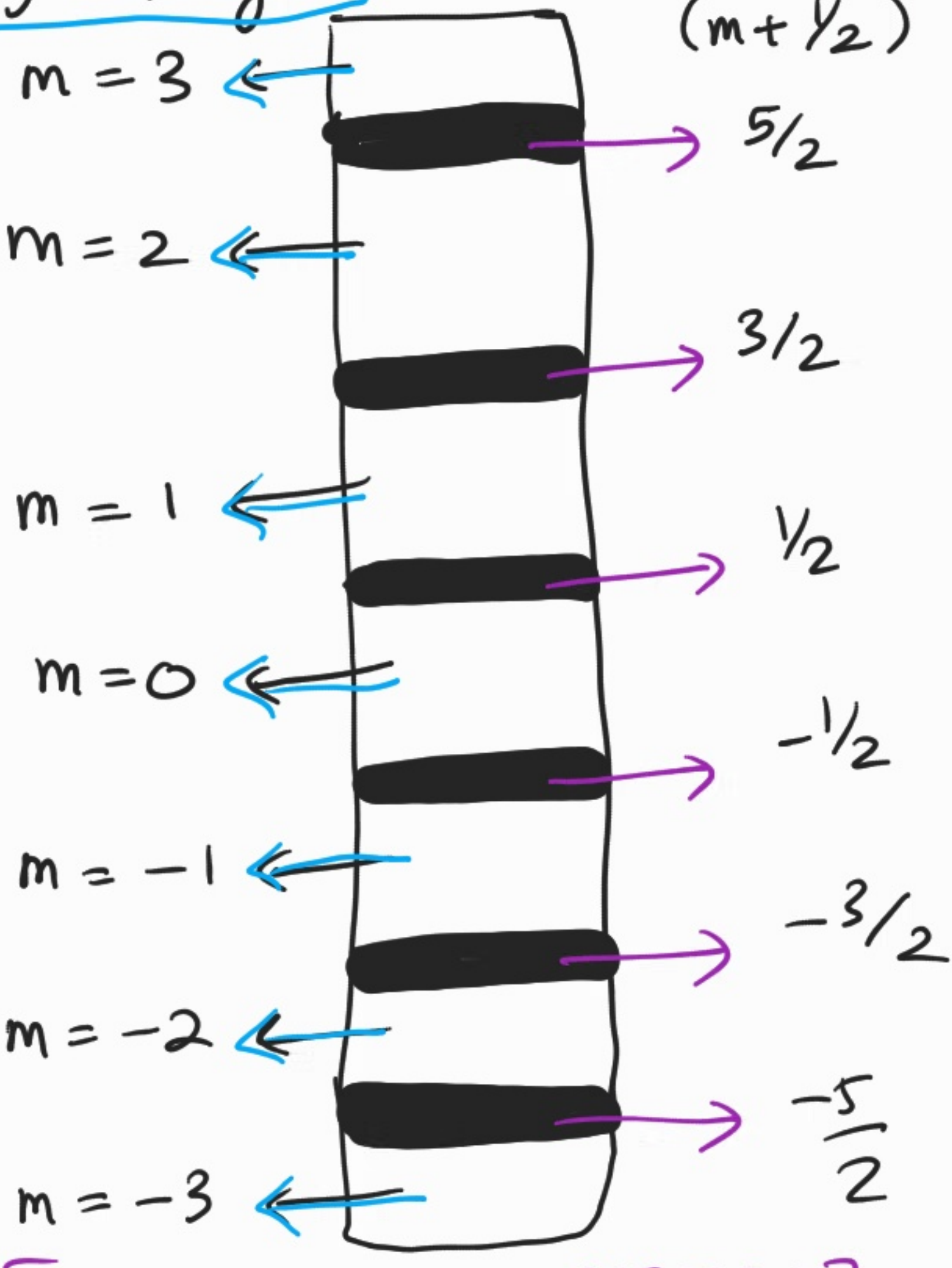


Young's Experiment

Bright fringes

Dark fringes

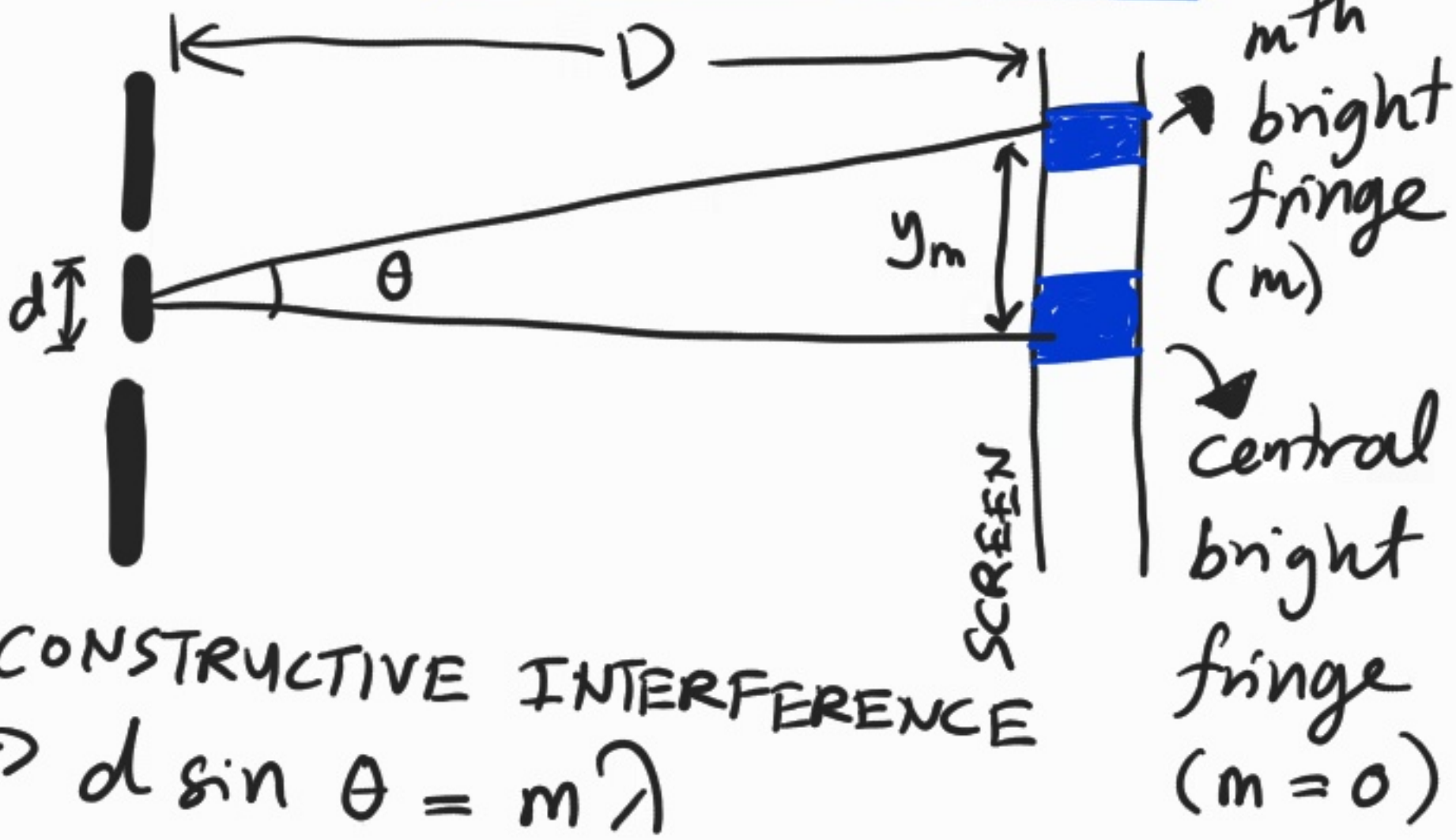
$(m + 1/2)$



[INTERFERENCE PATTERN]

Young's Experiment

y_m → the distance between the central bright fringe and the mth bright fringe.



CONSTRUCTIVE INTERFERENCE

↳ $d \sin\theta = m\lambda$

$\tan\theta = y_m / D$

⇛ for very small θ

$\tan\theta \approx \sin\theta$

$$\tan\theta = \frac{y_m}{D} \quad \text{——— (1)}$$

$$d\sin\theta = m\lambda$$

$$\sin\theta \approx \tan\theta \quad \text{(2)}$$

$$d\tan\theta = m\lambda \rightarrow \tan\theta = \frac{m\lambda}{d}$$

(1) = (2)

$$\frac{y_m}{D} = \frac{m\lambda}{d}$$

$$\boxed{y_m = \frac{m\lambda D}{d}}$$

Young's Experiment

Bright Fringes

$$y_m = \frac{mD\lambda}{d}$$

$\leftrightarrow$

Dark Fringes

$$y_m = \left(m + \frac{1}{2}\right)\frac{D\lambda}{d}$$

- The distance between 2 adjacent bright fringes (same for the dark ones):

$$y_{m+1} - y_m = (m+1)\frac{D\lambda}{d} - m\frac{D\lambda}{d}$$

$$\therefore \boxed{\Delta y = \frac{D\lambda}{d} \quad \text{at } m = 0}$$

Young's Experiment

- Intensity of the interference pattern with respect to phase difference:

$$I = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$I_0 \Rightarrow$ intensity of the light on the screen when one slit is covered. (no interference)

Young's Experiment

- Constructive Interference
  [Bright fringes]

$$\Delta\phi = 2m\pi$$

$$I = 4I_0 \cos^2(m\pi) = 4I_0$$

- Destructive Interference
  [Dark fringes]

$$\Delta\phi = (2m+1)\pi$$

$$I = 4I_0 \cos^2\left(m+\frac{1}{2}\right)\pi = 0$$

